# Adding and Subtracting in Scientific Notation

*Teacher's Edition*

## The Period

**MA.8.NSO.1.5** Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

**Must** Add and subtract numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

**Must not** The two exponents stay within 2 of each other — that is this benchmark's own limit. Multiplication and division are 4.02–03; significant digits are 4.04.

**Target** I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

**MTR** **MTR.5.1** — Notes I · **MTR.6.1** — Boards 3 and 4 · **MTR.4.1** — Board 6

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 11 min |
| Example 1 | 8–9 | 3 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 3 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 12 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. The powers of ten have to match before a single digit is added. That is the whole lesson.
2. The power of ten is a label, like the word thousands. You never add the label.
3. When you finish, look at the coefficient. If it is not between 1 and 10, you are not done.

## Slide by Slide

1. **Adding and Subtracting in Scientific Notation.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is today's lesson in miniature. *1 retrieves 3.09 — small numbers in scientific notation: the 6 is in the ten-thousandths place (yesterday). 2 retrieves 3.08 — divide the coefficients, divide the powers, multiply the results (last week). 3 retrieves Unit 2 — perfect squares (2.01). Roots come back for good in 4.05 (last unit). 4 retrieves MA.5.NSO.2.3 — adding decimals: line up the place values, not the digits (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. $6.2 \times 10^{-4}$ 2. 4,000 3. 12 4. 1.69
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the table.** You can only add things that are the same size. Ten-thousands add to ten-thousands, and the label never changes.
6. **Notes.** The power of ten is a label. Six thousands plus two thousands is eight thousands — you never add the word thousands.
7. **Notes.** The exponents are never added. Adding 10 to the fifth and 10 to the third does not make 10 to the eighth — it does not make a power of ten at all.
8. **Example 1 — Same power, and the answer is not finished.** SPOKEN ONLY.
9. **Example 1 — Worked.** Add the coefficients, keep $10^7$, then repair.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape with negative exponents.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Different powers, and a subtraction.** SPOKEN ONLY.
13. **Example 2 — Worked.** Rewrite $4.5 \times 10^5$ over $10^6$ — exponent up one, point one place left — and the subtraction becomes ordinary.
14. **Your Turn — Same steps, your numbers. Boards up when done.** The exponents are two apart — the widest gap this benchmark allows.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. **Board 1.** $(3.2\times10^5)+(4.1\times10^5)$ → ***7.3 × 10⁵*** *$7.3 \times 10^{10}$ — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]; $1.312 \times 10^{11}$ — the numbers were multiplied [8.NSO.1.5: the operation is addition]*

18–19. **Board 2.** A stadium sold $4.2 \times 10^4$ tickets on Friday and $3.5 \times 10^4$ on Saturday. Each ticket cost 38 dollars. How many tickets were sold on the two days? → ***7.7 × 10⁴*** *The ticket price of 38 dollars is not needed. anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]; $7.7 \times 10^8$ — the exponents were added [B1G-M 8.NSO.1.5 misconception]*

20–21. **Board 3.** $(8\times10^{-6})+(3\times10^{-6})$ → ***1.1 × 10⁻⁵*** *$11 \times 10^{-6}$ — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving $12 \times 10^9$ instead of $1.2 \times 10^{10}$]; $1.1 \times 10^{-11}$ — the exponents were added, giving $11 \times 10^{-12}$, and the answer was then finished*

22–23. **Board 4.** $(5.6\times10^8)-(4.9\times10^8)$ → ***7 × 10⁷*** *$0.7 \times 10^8$ — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]; $1.05 \times 10^9$ — added instead of subtracting*

**24–25. Board 5.** $(1.6 \times 10^4) + (2.3 \times 10^6)$. Which is the sum? → ***C — 2.316 × 10⁶***
*(A) the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]; (B) the coefficients were added and the exponents were added [B1G-M 8.NSO.1.5 named misconception: $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$]; (D) $1.6 \times 10^4$ was rewritten as $0.16 \times 10^6$ — the decimal point moved one place too few [8.NSO.1.5: the point moves as many places as the exponent changes] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**26–27. Board 6.** $(2.5 \times 10^6) + (9.8 \times 10^6)$. Which is the sum, written in scientific notation? → ***D — 1.23 × 10⁷*** *(A) equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving $12 \times 10^9$ instead of $1.2 \times 10^{10}$]; (B) the exponents were added as well as the coefficients [8.NSO.1.5: the power of ten does not change]; (C) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** $(7.8\times10^{-2})-(6.4\times10^{-3})$ → ***7.16 × 10⁻²*** *$1.4 \times 10^{-2}$ — the coefficients were subtracted without matching the powers [8.NSO.1.5]; $7.16 \times 10^{-5}$ — the exponents were added, $-2 + (-3)$ [8.NSO.1.5: the power of ten is a label, not a term]*

**30–31. Board 8.** $(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$. What is the value of n? → ***n = 5.7*** *−3.48 — the target was not rewritten over $10^6$ before subtracting [8.NSO.1.5: match the powers first]; $5.7 \times 10^6$ — the blank holds a coefficient, not a whole term*

**32–33. Board 9.** Devon writes $(1.9 \times 10^2) + (6.6 \times 10^4) = 8.5 \times 10^6$. Name the two mistakes he made, and give the correct sum. → ***6.619 × 10⁴*** *'he added wrong' — no mistake is actually named [the item asks which two]; $8.5 \times 10^4$ — the exponent mistake was caught and the matching mistake was not*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Questions 1–6 are one move each and walk the whole decision tree: same power with no repair, same power subtracting, a repair upward, a repair downward, a gap of 1, a gap of 2 with negative exponents. 7 and 8 are two real quantities each, one a sum and one a difference, so the same pair of numbers is used twice in different directions. 9 works backwards in three shapes — a missing addend, a missing subtrahend, and a sum that lands exactly on 10. The pattern breaks at 10, where the student reads someone else's work and has to name two mistakes rather than find one answer. 11 is the diagnostic multiple choice and 12 is the select-all, whose two wrong options are the two ways of misreading an exponent. The Additional sheet mirrors each position.

## Misconceptions to Watch

**Board 1.** $7.3 \times 10^{10}$ — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]

**Board 2.** anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]

**Board 3.** $11 \times 10^{-6}$ — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving $12 \times 10^9$ instead of $1.2 \times 10^{10}$]

**Board 4.** $0.7 \times 10^8$ — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]

**Board 5.** the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]

**Board 6.** equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving $12 \times 10^{9}$ instead of $1.2 \times 10^{10}$]

**Board 7.** $1.4 \times 10^{-2}$ — the coefficients were subtracted without matching the powers [8.NSO.1.5]

**Board 8.** −3.48 — the target was not rewritten over $10^{6}$ before subtracting [8.NSO.1.5: match the powers first]

**Board 9.** 'he added wrong' — no mistake is actually named [the item asks which two]

**Materials** Whiteboards and markers. Calculators are allowed on everything; entering 8E−6 + 3E−6 and reading 1.1E−5 back off the display is worth thirty seconds at whiteboard 3.